

Build a highly scalable, single-file monolithic financial SaaS application, public investment community, and educational platform in Node.js/Express with an embedded React frontend (via CDN) and a persistent SQLite database using `sqlite3` or `better-sqlite3`. Name the main file `index.js`. Brand the platform as "Sievoos" (Domain: Sievoos.com). The app operates as an autonomous quantitative portfolio management engine, Discounted Cash Flow (DCF) valuation suite, FIRE/Pension calculator, public analysis sharing feed, and content portal designed for Google AdSense monetization and high organic SEO traffic.

BRAND IDENTITY, LOGO & DESIGN SYSTEM FOR SIEVOOS.COM

1. Brand Identity: "Sievoos.com" - The Intelligent Quality Sieve & Valuation Engine for Investors.
2. Visual Theme & Aesthetics: Ultra-sleek, modern dark-mode FinTech aesthetic.
 - Primary Background: Deep Charcoal / Slate (`#0b0f19`).
 - Card Backgrounds: Dark Slate (`#1e293b`).
 - Accent 1 (Sieve Gold): Bright Amber/Gold (`#f59e0b`) symbolizing elite quality.
 - Accent 2 (Growth Emerald): Vibrant Emerald (`#10b981`) symbolizing valuation upside.
 - Accent 3 (Crisis Rose): Rose/Red (`#f43f5e`) for risk floors and gatekeeper failure alerts.
3. SVG Logo & Favicon: Render a custom inline SVG logo featuring a stylized geometric "S" shield combined with a fine-mesh sieve pattern and an ascending growth curve. Include an embedded SVG Favicon in the HTML `` so browser tabs display the Sievoos icon seamlessly.

THE COMPLETE INVESTMENT SYSTEM & PORTFOLIO ALLOCATION ENGINE

MASTER FORMULA FOR PORTFOLIO ALLOCATION WEIGHT (W_final):

$$W_{final} = (B_{NoBS} * B_{Moat} * B_{CEO} * B_{DCFbear}) * \max(0.05, \min(0.10, ((0.30 * S_{Rule40} + 0.30 * S_{DCFBase} + 0.25 * S_{Insider} + 0.15 * S_{Thesis}) / \beta) * 0.22))$$

[PENSION & FIRE GOAL ENGINE: THE 25X RULE & 4% WITHDRAWAL RULE]

Integrate a dedicated Retirement / Financial Independence Target Engine based on the 4% Rule and 25x Annual Expenses formula:

1. The 25x Expenses Rule: The total capital required for full retirement/financial freedom is calculated dynamically as:

Required Target Capital = Desired Annual Expenses * 25

2. The 4% Rule Explanation & Mechanics: Include an interactive educational component explaining the Trinity Study 4% Safe Withdrawal Rate (SWR). It illustrates how withdrawing 4% of total portfolio value in Year 1 (adjusted for inflation) allows capital to last 30+ years when allocated according to the Core-Satellite Rule of 110 model.

3. Gap & Runway Calculator: Compare current capital vs Required Target Capital, calculating remaining gap and projected target completion year based on monthly savings.

[CORE ARCHITECTURE & ASSET ALLOCATION (RULE OF 110)]

1. Core-Satellite Discipline: Satellite individual stocks strictly kept to 3 to 10 positions (ideal 3-5). Broad market indexing serves as the psychological anchor.

2. Asset Allocation Matrix (Rule of 110):

- Target Horizon <= 5 Years: 70% Defense (Bonds/Fixed Income) | Remaining equities split 50/50 Core/Satellite.

- Target Horizon <= 10 Years: 50% Defense | Remaining equities split 50/50 Core/Satellite.

- Target Horizon <= 25 Years: 20% Defense | Remaining equities split 50/50 Core/Satellite.

- Target Horizon > 35 Years: 10% Defense | Remaining equities split 50/50 Core/Satellite.

- Fallback Redirect: If fewer than 3 individual stocks pass the sieve, automatically redirect satellite capital into a technology index ETF (e.g., QQQM).

MODULE 1: THE DISCOUNTED CASH FLOW (DCF) VALUATION ENGINE

Before running the Master Formula, the system automatically calculates the intrinsic value per share under three scenarios (Bear Case, Base Case, Bull Case) using the following 4-step DCF methodology:

STEP 1: CALCULATE THE DISCOUNT RATE / COST OF CAPITAL (r or WACC)

First, calculate the required Return on Equity (Re) using the Capital Asset Pricing Model

(CAPM):

$$R_e = R_f + \text{Beta} * (R_m - R_f)$$

- * R_f (Risk-Free Rate): Yield on the 10-year U.S. Treasury bond.

- * Beta: Stock volatility relative to the broader market.

- * $(R_m - R_f)$ (Market Risk Premium): Extra return demanded above risk-free investments (typically 6%, assuming $R_m = 10\%$ and $R_f = 4\%$).

When a firm carries debt, calculate the Weighted Average Cost of Capital (WACC) to use as the discount rate (r):

$$\text{WACC} = (E / V * R_e) + (D / V * R_d * (1 - T_c))$$

Where E is market value of equity, D is market value of debt, V is $E + D$, R_d is cost of debt, and T_c is corporate tax rate.

STEP 2: PROJECT AND DISCOUNT FREE CASH FLOWS (FCF FOR YEARS 1 TO 5)

Project future revenues for Years 1 through 5 applying a safety margin cushion (e.g., use 10% expected growth instead of 12%). Multiply projected revenue by the company's 3-year average Free Cash Flow (FCF) margin to get FCF for each year:

$$\text{FCF}_t = \text{Revenue}_t * \text{FCF_Margin}$$

Discount each year's FCF back to present value (PV) using the discount rate (r):

$$\text{PV}_t = \text{FCF}_t / (1 + r)^t$$

$$\text{PV_Sum} = \text{Sum of PV}_1 \text{ through PV}_5$$

STEP 3: CALCULATE TERMINAL VALUE (TV) AND PRESENT VALUE OF TV (PV_TV)

Calculate the value of the business beyond Year 5 using either the Perpetuity Approach or the EBITDA Multiple Approach:

- * Perpetuity Growth Approach: $\text{TV} = (\text{FCF}_5 * (1 + g)) / (r - g)$ [where g is long-term sustainable growth, e.g., 2% to 4%]

- * EBITDA Multiple Approach: $\text{TV} = \text{EBITDA}_5 * \text{EV_EBITDA_Multiple}$

Discount Terminal Value back to present value:

$$\text{PV_TV} = \text{TV} / (1 + r)^5$$

STEP 4: CALCULATE INTRINSIC EQUITY VALUE PER SHARE (V_DCF)

- * Enterprise Value (EV) = $\text{PV_Sum} + \text{PV_TV}$

- * Equity Value = Enterprise Value + Cash and Cash Equivalents - Total Debt

- * Intrinsic Value per Share (V_{DCF}) = Equity Value / Total Shares Outstanding

DCF SCENARIOS PRODUCED:

- * $V_{\text{DCF_Bear}}$ (Bear Case DCF): Conservative assumptions (lower revenue growth, reduced margins, higher discount rate $r + 1\%$).

- * $V_{\text{DCF_Base}}$ (Base Case DCF): Realistic assumptions using 3-year historical averages with safety cushions.

- * $V_{\text{DCF_Bull}}$ (Bull Case DCF): Optimistic assumptions based on company guidance.

=====

MODULE 2: BREAKDOWN OF THE MASTER FORMULA (FROM LEFT TO RIGHT)

THE BINARY QUALITY SIEVE (Left Side):

This acts as a strict, zero-tolerance gatekeeper. Each binary test returns either 1 (Pass) or 0 (Fail). If a single test returns 0, the entire left side becomes 0. Because 0 multiplied by anything equals 0, the formula instantly sets $W_{\text{final}} = 0\%$. The stock is disqualified and must not be owned.

- * B_{NoBS} : Set to 1 if the company meets at least 2 out of 3 tests: Actual 3-year historical revenue growth (not forecasts), positive Free Cash Flow (not adjusted EBITDA), and more cash than long-term debt on the balance sheet. Otherwise 0.

- * B_{Moat} : Set to 1 if the company possesses an identifiable competitive advantage (Brand, Network Effects, High Switching Costs, Data Moat, Cost Advantage, or Regulatory Approvals) confirmed by a stable or expanding operating margin over time. Otherwise 0.

- * B_{CEO} : Set to 1 if the CEO/management has genuine "Skin in the game" through open-market stock purchases (not just option grants), a strong execution track record, or founder status / 10+ years tenure. Otherwise 0.

- * $B_{\text{DCF_Bear}}$: Set to 1 if the current real-market stock price is lower than the calculated conservative Bear Case DCF value ($\text{Current Price} < V_{\text{DCF_Bear}}$). Otherwise 0.

THE RISK-ADJUSTED ALLOCATION ENGINE (Right Side):

If the stock passes all binary gates (left side = 1), the right side activates to calculate the exact target position size as a percentage of your total net worth:

- * S_{Rule40} (Growth + Margin Score): Score = Revenue Growth Rate % + Free Cash Flow Margin %. If Score $\geq 100\%$, award 100 points (Elite). If Score $\geq 40\%$, award 60 points (Baseline). If Score $< 40\%$, award 0 points.

- * $S_{\text{DCF_Base}}$ (Valuation Score): Measures the percentage upside from the current market

price up to the Base Case DCF value (V_DCF_Base). Higher discounts to intrinsic value generate higher scores (capped at 100).

* S_Insider (Insider Activity Score): Score from 0 to 100 based on the volume, frequency, and role (CEO/CFO vs. Board) of open-market insider purchases filed via official regulatory disclosures over the past 6 months.

* S_Thesis (Market Misunderstanding Score): Manual evaluation from 0 to 100 rating how severely the broader market misunderstands the company's real business transformation. Default baseline is 50.

* Beta (Volatility Factor): Historical stock volatility relative to the market index. Because the total weighted score is divided by Beta, highly volatile stocks (high Beta) are automatically scaled down in position size to protect the portfolio against severe drawdowns.

* 0.22 (Scaling Constant): A mathematical normalizing constant used to translate raw points into correct target percentage weights.

* max(0.05, min(0.10, ...)) (Safety Cap Lock): Ensures that regardless of how high or low a qualified stock scores, its final target position size will never exceed 10% (preventing concentration risk and "Danger Zone" overexposure) and will never fall below 5% (the minimum threshold for a satellite stock).

=====

MODULE 3: ROUTINE EXECUTION & ACTIONABLE DECISION MATRIX

Every Sunday evening, compare the Target Weight generated by the formula (W_final) against the actual current weight of the stock in your total portfolio (W_actual):

* DECISION 1: If W_final = 0%

Action: EXCLUSION OR FULL EXIT. Do not open a position. If currently owned, sell 100% of the position immediately because the investment thesis or quality criteria have failed.

* DECISION 2: If W_actual is LESS than W_final (Underweight by > 2%)

Action: HOLD + ADD. The stock is underweight relative to its target. This stock becomes your primary target for regular Dollar-Cost Averaging (DCA). Furthermore, if the stock price drops 20% or more below its 52-week high, trigger the "Double-Down Rule" and deploy 3X your normal DCA amount using cash from your Dry Powder Bank.

* DECISION 3: If W_actual is GREATER than W_final (Overweight by > 2%)

Action: HOLD + TRIM. The stock has run up significantly and now creates unapproved concentration risk. Surgically shave off 10% to 20% of the position to bring its weight back down to W_final. Reallocate the proceeds directly into your Core index funds (e.g., S&P 500) or High-Yield Dry Powder Bank.

=====

[MODULE 4: PERSISTENT SQLITE DATABASE ARCHITECTURE]

Use SQLite ('sqlite3' / 'better-sqlite3') to permanently persist data across server restarts:

1. 'contact_messages': (id, name, email, subject, message, created_at, read_status).
2. 'admin_config': (id, password_hash, session_token).
3. 'published_analyses': (id, title, ticker, current_price, base_dcf, bear_dcf, margin_of_safety, user_notes, full_inputs_json, likes_count, created_at, author_alias, edit_pin).
4. 'comments': (id, analysis_id, author_name, comment_text, created_at).

=====

[MODULE 5: FRICTIONLESS COMMUNITY FEED, COMMENTS & "FORK" FEATURE]

1. Public Community Feed ('/community' or Home tab): Displays user-submitted stock analyses. Includes sorting parameters: Most Liked, Newest, Highest Margin of Safety, and Ticker Search.
2. Frictionless Publishing & Commenting: Users publish analyses or write comments by entering a custom Display Alias (stored in LocalStorage) without registration. Include optional 4-digit PIN for post editing.
3. "Copy/Fork to My Calculator" Button: Clicking loads raw valuation parameters from shared posts directly into the active calculator with 1 click so users can tweak assumptions effortlessly without manual data entry.

=====

[MODULE 6: SIEVOO DIAGNOSTIC ERROR ENGINE & SOCIAL SHARE CARDS]

1. Binary Gate Diagnostic Checklist: When W_final calculates to 0%, display a prominent visual checklist showing exactly which gate failed (B_NoBS, B_Moat, B_CEO, B_DCFBear) and why.
2. HTML5 Canvas Shareable Card Generator: "Export as Image/Share Card" button dynamically draws a dark-mode summary graphic branded with Sievoo.com containing Ticker, Current Price vs Fair Value, Margin of Safety %, Action Directive, and Sievoo logo. Includes text-copy formatted for Twitter/X, Reddit, and Discord.

=====

[MODULE 7: EDUCATIONAL ARTICLES & SEO HUB (ADSENSE MANDATE)]

Include a dedicated "Sievo Academy" tab containing 4 comprehensive SEO-optimized guide articles:

1. "Understanding Discounted Cash Flow (DCF): WACC, FCF & Terminal Value Explained."
2. "The Core-Satellite Strategy and Rule of 110 Asset Allocation."
3. "The 4% Rule and 25x Expenses Formula for Pension and FIRE."
4. "Evaluating Moats, Rule of 40, and Insider Activity in Growth Stocks."

=====

[MODULE 8: BACKEND CONTACT SYSTEM & PROTECTED ADMIN CONSOLE]

1. Endpoint `POST /api/contact`: Public message submission (Name, Email, Subject, Message) saved securely to SQLite.
2. Endpoint `/admin` Route & API Console: Protected Dashboard.
 - Password Authentication: Default initial password (`AdminPass123!`).
 - Message Management: View, mark as read, or delete contact messages stored in SQLite.
 - Admin Password Management: Generate and update new admin password stored securely in SQLite.

=====

[MODULE 9: LEGAL MANDATES, DATA PORTABILITY & ADSENSE LAYOUT]

1. Legal Pages & Footer Links: Dedicated Privacy Policy, Terms of Service, About Us, and Financial Disclaimer ("FOR EDUCATIONAL AND INFORMATIONAL PURPOSES ONLY. NOT FINANCIAL, LEGAL, OR TAX ADVICE. Conduct your own due diligence (DYOR). Never follow social media influencers blindly.") in footer for 100% Google AdSense approval.
2. Data Portability: "Download Portfolio Report (.json)" button, "Upload Saved Report" file picker, and printer-friendly PDF view.
3. Anti-Spam Rate Limiting: Basic IP throttling on contact and comment endpoints to prevent spam.
4. Ads.txt Endpoint: Include express route `/ads.txt` ready for Google AdSense integration.
5. Localization Engine: Seamless toggle between Norwegian (`no`) and English (`en`) without state loss across all UI elements, legal pages, and documentation text.
6. Ad Placement Zones: Visible layout wrappers `[REKLAMEANNONSE (HÅY CTR SONE)]` at top banner, inner stream context, and footer grid.

Ensure the entire architecture is packaged inside a single executable `index.js` container running Node.js/Express with SQLite. Generate production-ready code.